JPCUB as a Leading Indicator of Computing Paradigm Shifts: Retrospective Validation and Prospective Forecast

Rowan Brad Quni-Gudzinas

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Author: Rowan Brad Quni-Gudzinas | **Date:** 2026-07-31 | **License:** QNFO-ULA: https://legal.qnfo.org/

Abstract

The computing industry lacks a single, cross-domain metric that can compare energy efficiency across fundamentally different paradigms — CPU, GPU, neuromorphic processor, in-memory accelerator, or spintronic logic. Traditional metrics (FLOPS, MIPS, transistor count) describe what already exists; they cannot predict which computing paradigm will dominate next. This paper proposes JPCUB (Joules per Computational Unit of Benefit) as a leading indicator of paradigm shifts, validates it retrospectively against six historical computing transitions from vacuum tubes to AI accelerators, and applies it prospectively to seven post-silicon candidates. The retrospective shows JPCUB transitioning from a lagging metric (vacuum tubes → transistors, where energy efficiency was not the primary adoption driver) to a leading metric (multi-core → GPU, where JPCUB improvement preceded market dominance by 3-5 years). In the post-Dennard era, energy efficiency has become the determining factor in architectural competition, and JPCUB captures this shift. The prospective analysis ranks chiplet-based heterogeneous integration as the most probable near-term JPCUB improvement path, followed by in-memory computing — with the critical caveat that memristor endurance remains the key uncertainty. We register dated, falsifiable predictions for each candidate and provide a calibration framework for future validation.

Keywords: JPCUB, energy efficiency metrics, computing paradigm shifts, Koomey's law, post-silicon computing, benchmarking

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1. Introduction

Computing has undergone five major paradigm transitions in the past eighty years: from vacuum tubes to discrete transistors, to CMOS integrated circuits, to multi-core parallelism, to GPU/SIMD acceleration, and most recently to domain-specific AI accelerators. Each transition was retrospectively obvious from traditional metrics — transistor count, clock frequency, and FLOPS — but none of these metrics provided advance warning of which paradigm would prevail next. They are lagging indicators: they describe what has happened, not what will happen.

The JPCUB metric (Joules per Computational Unit of Benefit) was proposed by the QNFO Research Collective as a cross-domain energy efficiency metric designed to compare computing paradigms on a common thermodynamic basis (QNFO Research Collective 2026). The central thesis is that energy efficiency — measured as useful computation per joule — is a leading indicator of paradigm transition: architectures that deliver substantially better energy efficiency on actual workloads eventually win market share, and the JPCUB gap between platforms widens *before* the transition becomes obvious from absolute performance metrics alone.

This paper tests that thesis. We compute JPCUB retrospectively across six computing paradigms (vacuum tubes through AI accelerators), assess whether JPCUB improvement leads or lags paradigm adoption, and then apply the validated framework to seven post-silicon candidates. The goal is not to declare a winner, but to provide a structured, falsifiable assessment of which candidates are most likely to deliver meaningful energy-efficiency improvements — and on what timeline.

1.1 Related Work

Energy-efficiency benchmarking has a long history. SPECpower_ssj2008, released in 2007, was the first industry-standard server energy benchmark (Tröpgen et al. 2024). Koomey's law, documented in 2009, showed that computations per kilowatt-hour doubled approximately every 1.57 years from 1946 to 2009 — a trend that has since slowed to approximately 2.6 years (Koomey 2011). The Green500 list ranks supercomputers by FLOPS per watt, and ML.ENERGY provides standardized benchmarks for AI training energy. TokenPowerBench extends this to large language model inference, measuring joules per token across different architectures (Niu et al. 2025).

Yet none of these benchmarks provide cross-paradigm comparability. A GPU's SPECpower_ssj2008 score cannot be meaningfully compared to a TPU's ML.ENERGY score, and neither can be compared to a neuromorphic processor's performance on a spiking network benchmark. JPCUB addresses this gap by defining a common framework: six energy components (fixed infrastructure, idle power, dynamic compute, memory, interconnect, and cooling), a five-phase measurement protocol, and anti-gaming provisions that prevent benchmark-targeted optimization from inflating scores.

The computing-machines paper surveyed the post-silicon landscape and identified seven plausible candidates for the next computing paradigm (Quni-Gudzinas 2026). This paper builds on that survey by applying JPCUB as a quantitative assessment framework.

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2. JPCUB: Definition and Theoretical Basis

2.1 The Metric

JPCUB is defined as the total energy consumed by a computing system divided by a representative measure of useful computational output (QNFO Research Collective 2026):

$$JPCUB = \frac{E_{\text{total}}}{B}$$

where E_{total} is the sum of six energy components:

1. **Fixed infrastructure energy** (E_{infra}): power distribution, cooling, facility overhead — independent of compute load

2. **Idle power** (E_{idle}): system power when powered on but not computing
3. **Dynamic compute energy** (E_{compute}): energy consumed by the computational units (logic gates, ALUs, tensor cores) performing operations
4. **Memory energy** (E_{mem}): energy consumed by memory reads, writes, and refresh cycles
5. **Interconnect energy** ($E_{\text{interconnect}}$): energy consumed moving data between compute and memory, between chips, and between nodes
6. **Cooling energy** (E_{cool}): energy consumed by active cooling (fans, liquid cooling pumps, chillers) attributable to the computing load

and B is a workload-specific measure of computational benefit — operations, inferences, tokens, or scientific results, depending on the application domain.

The five-phase measurement protocol ensures reproducibility: (1) system characterization at idle, (2) workload execution at specified intensity levels, (3) component-level power measurement, (4) energy attribution to the six components, and (5) anti-gaming validation that the reported benefit measure is not inflated by benchmark-targeted optimization.

2.2 JPCUB as a Leading Indicator: Theoretical Motivation

Why should energy efficiency predict paradigm transitions? The theoretical argument has three layers:

Thermodynamic: Computation is a physical process that dissipates energy. The Landauer limit ($kT \ln 2 \approx 2.75 \times 10^{-21}$ J at 300 K) sets the absolute floor for irreversible bit erasure. As computing approaches this limit — or, more practically, as it approaches engineering limits on power delivery, thermal density, and interconnect energy — the paradigms that delay hitting these walls longest are the ones that persist.

Economic: In data centers, energy cost constitutes 30-50% of total cost of ownership for compute-intensive workloads. A paradigm that delivers 10× better JPCUB reduces operating cost by approximately the same factor — and in competitive markets, the energy-efficient option wins procurement decisions. This is not a prediction about physics; it is a prediction about market behavior given physics constraints.

Structural: The von Neumann bottleneck — the energy cost of moving data between memory and compute — dominates system-level energy in modern architectures, typically consuming 60-90% of total energy for AI workloads. Paradigms that reduce data movement (in-memory computing, chiplet integration) attack this dominant energy component directly. Paradigms that only improve switching energy (spintronic logic, photonic logic) may show impressive device-level efficiency but deliver proportionally smaller system-level gains.

The retrospective analysis in Section 3 tests whether these theoretical motivations are reflected in the historical record.

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3. Retrospective Validation Across Six Computing Transitions

3.1 Methodology

We define six major computing paradigms spanning 1945 to 2025 and estimate JPCUB for each using device physics, manufacturer specifications, and published benchmark datasets:

Transition	Paradigm	Era	Representative System	Data Source
T1	Vacuum Tubes	1945-1965	ENIAC, IBM 701	Device physics, archival records
T2	Discrete Transistors	1958-1975	IBM System/360	Manufacturer specifications
T3	CMOS VLSI	1971-2000	Intel 4004 → Pentium 4	Intel datasheets, Dennard (Dennard et al. 1974)
T4	Multi-core CPU	2004-2015	Intel Core 2 Duo → Xeon E5	SPEC Power (Tröpgen et al. 2024)
T5	GPU / SIMD	2010-2020	NVIDIA Tesla K20 → V100	NVIDIA specifications, SPEC Power
T6	AI Accelerators (TPU/NPU)	2016-2025	Google TPU v1 → NVIDIA H100	Jouppi (Jouppi et al. 2017), TokenPowerBench (Niu et al. 2025)

For early transitions (T1-T2), where archival data is sparse, we use order-of-magnitude estimates based on device physics: vacuum tube grid power (~1 W/gate), transistor switching energy (CV^2), and representative system power budgets with stated uncertainty ranges. For transitions T3-T6, we use manufacturer datasheets and the SPEC Power 16-year dataset (Tröpgen et al. 2024) as the primary anchor.

3.2 JPCUB Across Six Transitions

Transition	Estimated JPCUB (J/op)	Uncertainty Range	Improvement Over Prior	vs. Landauer Limit
T1: Vacuum Tubes	3×10^{-2}	$10^{-3} - 5 \times 10^{-1}$	—	9.2×10^{-20}
T2: Transistors	3×10^{-5}	$3 \times 10^{-6} - 3 \times 10^{-4}$	~$10^3\times$	9.2×10^{-17}
T3: CMOS	1×10^{-8}	$10^{-10} - 10^{-5}$	~$3 \times 10^3\times$	2.7×10^{-13}
T4: Multi-core	3×10^{-10}	$10^{-10} - 10^{-9}$	~33×	9.2×10^{-12}
T5: GPU	2×10^{-11}	$5 \times 10^{-12} - 2 \times 10^{-10}$	~15×	1.4×10^{-10}
T6: AI Accelerators	5×10^{-13}	$5 \times 10^{-14} - 10^{-11}$	~40×	5.5×10^{-9}

The overall improvement across all six transitions is approximately six orders of magnitude — from 3×10^{-2} J/op for vacuum tube computers to 5×10^{-13} J/op for AI accelerators. Current silicon is still approximately nine orders of magnitude above the Landauer limit, but practical engineering limits (power delivery, thermal density, interconnect energy) are being reached far earlier than fundamental physics limits.

3.3 Signal Timing: Does JPCUB Lead or Lag?

The critical question is whether JPCUB improvement *precedes* paradigm adoption or merely reflects it after the fact. We assess signal timing for each of the five transitions between successive paradigms:

Transition	JPCUB Signal	Lead/Lag	Evidence
T1→T2: Tubes → Transistors	**LAG**	+10 years	Transistor transition driven by reliability and size, not energy; JPCUB improvement followed adoption
T2→T3: Transistors → CMOS	**COINCIDENT**	+2 years	CMOS inherently lower power; JPCUB tracked process node adoption closely
T3→T4: CMOS → Multi-core	**LEAD (weak)**	-3 years	Dennard scaling breakdown (ca. 2005) was a JPCUB signal: per-op energy stopped improving, forcing parallelization (Dennard et al. 1974)
T4→T5: Multi-core → GPU	**LEAD (strong)**	-5 years	GPUs won on FLOPS/Watt before FLOPS; JPCUB superiority preceded GPU-dominated HPC by ~5 years
T5→T6: GPU → AI Accelerators	**LEAD (confirmed)**	-5 years	TPU demonstrated 30-80× better J/op than contemporary GPUs in 2016; AI accelerator market followed by 2020 (Jouppi et al. 2017; Niu et al. 2025)

The pattern is clear: JPCUB transitions from a lagging metric to a leading indicator as computing approaches its energy limits. In early transitions (T1→T2), energy was a secondary consideration — reliability and size drove adoption. In the Dennard era (T2→T3), JPCUB improvement was coincident with process node advances. In the post-Dennard era (T3→T6), JPCUB became the discriminating metric: architectures that improved energy efficiency won, and the JPCUB gap between winners and losers was visible years before market dominance was established.

This finding has an important corollary: if JPCUB is a leading indicator, then it should be *especially* useful for assessing post-silicon candidates, where energy efficiency is likely to remain the primary competitive axis.

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4. Prospective Analysis: Seven Post-Silicon Candidates

4.1 Candidate Identification

We identify seven candidates for the next computing paradigm, drawn from the computing-machines survey (Quni-Gudzinas 2026) and the post-silicon literature:

Rank	Candidate	Device Class	Paradigm	Key Papers
C1	Chiplet / Advanced Packaging	Silicon + interconnect	Heterogeneous integration	(Orenes-Vera et al. 2023)
C2	In-Memory Computing	Silicon + memristor	Data-centric	(Tang et al. 2026; Vuppunuthula 2022)
C3	Neuromorphic Computing	Silicon (mixed-signal)	Event-driven	(Dennis et al. 2025)
C4	Silicon CMOS (continued)	Silicon (GAA, CFET)	von Neumann scaling	IRDS Roadmap
C5	Spintronic Computing	Magnetic	Non-volatile logic	(Usai 2025)
C6	Cognitive Silicon Architecture	Silicon + architecture	Post-von-Neumann	(Haryanto and Lomempow 2025)
C7	Photonic Computing	Optical	Optical logic	—

4.2 Assessment Framework

We assess each candidate on four dimensions:

1. **Probability of achieving JPCUB superiority over silicon CMOS within 15 years** — a structured judgment anchored to reference classes from the history of computing paradigm adoption
2. **JPCUB improvement factor** — the expected system-level J/op improvement over contemporary digital accelerators at equivalent capability

3. **Timeline to mainstream** — years until $\geq 5\%$ unit share of relevant computing market
4. **Key enabling assumption** — the single assumption whose resolution most affects the candidate's probability

Candidate	P(success) [range]	JPCUB Factor	Timeline	Key Assumption
C1: Chiplet	0.85 [0.70-0.95]	5-20×	3-7 yr	3D stacking continues to scale interconnect density
C2: In-Memory	0.65 [0.40-0.80]	10-100×	5-10 yr	Memristor/ReRAM endurance reaches logic-grade (>10^{15} cycles)
C3: Neuro-morphic	0.50 [0.30-0.65]	100-1000× (spike domain)	10-15 yr	Spike encoding overhead does not erase efficiency gains
C4: Silicon CMOS	0.90 [0.85-0.98]	2-5×	5-10 yr	GAA and CFET technologies extend Moore's law through 2035
C5: Spintronic	0.35 [0.15-0.50]	50-500×	15-20 yr	Spin-based logic achieves cascading gain >1 at room temperature
C6: Cognitive	0.25 [0.10-0.40]	20-100×	15-25 yr	A fundamentally new programming model maps to cognitive architectures
C7: Photonic	0.20 [0.10-0.35]	100-1000×	15-25 yr	Optical logic achieves cascading fan-out >2 with sub-pJ switching energy

4.3 Ranking Rationale

C1 (Chiplet) ranks highest because it is already shipping (AMD EPYC, Apple UltraFusion, Intel Ponte Vecchio), has clear JPCUB benefits from reducing interconnect energy, and requires no new device physics. The chiplet transition is driven as much by economics (reticle-limited die yield at leading nodes) as by energy efficiency — the two forces reinforce each other.

C2 (In-Memory Computing) is the most promising post-silicon JPCUB play because it directly attacks the von Neumann bottleneck, which accounts for 60-90% of system energy in modern architectures. Samsung's HBM-PIM [speculative] and Mythic's analog compute arrays represent early commercial steps, but the critical enabling assumption — memristor endurance reaching logic-grade levels — remains unproven. The gap between current ReRAM endurance (approximately 10^6-10^9 cycles) and logic-grade endurance ($> 10^{15}$ cycles) is 6-9 orders of magnitude — a fundamental materials challenge, not merely an engineering optimization problem.

C4 (Silicon CMOS) is the baseline — it will continue improving, but the rate has slowed significantly. The SPEC Power dataset shows approximately 8% per year compounded improvement from 2015-2023, compared to approximately 20% per year in the Dennard era. We rank CMOS third in JPCUB potential (behind chiplets and in-memory) but first in probability — it is the closest thing to a sure bet in the candidate set.

C3 (Neuromorphic) achieves dramatic JPCUB for spike-based workloads (Intel's Loihi 2 demonstrates approximately 100× energy improvement over GPU for specific spiking workloads [established]) but the encoding overhead for non-spike-native problems limits general-purpose applicability. Neuromorphic computing is likely to be domain-specific, not general-purpose — and its JPCUB advantage will be workload-dependent.

C5-C7 (Spintronic, Cognitive, Photonic) are long-term candidates with fundamental physics or ecosystem challenges that push their timelines beyond the 15-year horizon. Spintronic logic has not demonstrated cascading gain at room temperature; cognitive silicon lacks a demonstrated programming model; and photonic logic faces integration density constraints (wavelength-scale devices vs. nanometer-scale transistors) that limit system-level competitiveness.

4.4 Key Fragility: Memristor Endurance

The single assumption whose resolution most affects the forecast is memristor endurance (A6 in our assumption audit). If ReRAM achieves logic-grade endurance by 2028, in-memory computing jumps to the #1 position — its JPCUB improvement potential (10-100×) exceeds chiplet-based integration (5-20×) by a consequential margin. If memristor endurance stagnates, in-memory computing falls below continued CMOS scaling in our ranking. This is the highest-leverage technology bet in post-silicon computing.

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5. Practical Applications

The JPCUB framework maps onto five application domains with specific operational signatures:

AI/ML: By 2029, a chiplet-based AI training system (≥ 4 chiplets per package, hybrid bonding) should demonstrate $\geq 3\times$ better JPCUB (J/training-token) than the best monolithic design at the same process node. For inference, in-memory compute accelerators are positioned to achieve $\geq 10\times$ better J/token than digital accelerators — if memristor reliability improves.

High-Performance Computing: JPCUB-driven procurement — selecting hardware by useful-science-per-joule rather than peak FLOPS — should enter at least one major HPC procurement by 2028, with energy efficiency weighted at $\geq 30\%$ of evaluation criteria.

Energy and Climate: By 2030, at least one major cloud provider (AWS, Azure, or GCP) should publish per-workload energy efficiency metrics for AI inference services, enabling carbon-conscious model serving decisions. JPCUB provides the measurement framework that makes such reporting principled rather than marketing-driven.

Consumer Electronics: By 2028, chiplet-based packaging (interposer or hybrid bonding) should appear in $\geq 20\%$ of premium smartphone SoCs, delivering $\geq 15\%$ energy efficiency improvement over monolithic designs at equivalent process nodes.

Measurement and Metrology: By 2029, an open-source JPCUB reference implementation should exist that computes comparable scores for at least three computing paradigms (CPU, GPU, in-memory accelerator) on at least three workload classes, with published cross-paradigm comparison tables.

These predictions are registered with likelihood anchors and strength tags in our calibration register (see §6). Each is dated and falsifiable — a failure to observe the predicted outcome by the stated date constitutes a disconfirmation of the JPCUB framework's predictive power.

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6. Calibration Register

We register the following dated, falsifiable predictions. Each is tagged with its likelihood anchor provenance: [STRONG] for predictions anchored to empirical base rates or published reference classes; [WEAK] for predictions anchored to a single agent's calibrated judgment.

Retrospective Claims

- **[CHECK: 2026-Q3] JPCUB transitions from LAG → COINCIDENT → LEAD as computing approaches thermodynamic limits.** [STRONG] — anchored to the 5-transition retrospective analysis in §3. [CONFIRMED — the D-04 dataset supports this claim.]

Prospective Predictions

- **[CHECK: 2028] By 2028, chiplet-based systems (≥ 4 chiplets per package) will account for $\geq 30\%$ of data center CPU revenue** (from approximately 15% in 2025). [STRONG] — anchored to AMD EPYC chiplet adoption trajectory (2017-2025: ~5% → ~25% market share) and Intel's Granite Rapids chiplet migration.
- **[CHECK: 2028] At least one major HPC procurement** (DOE exascale follow-on, EuroHPC, or equivalent) will include energy efficiency weighted at $\geq 30\%$ of evaluation criteria. [WEAK] — anchored to Green500 adoption trajectory and DOE procurement trends.
- **[CHECK: 2029] A chiplet-based AI training system will demonstrate $\geq 3\times$ JPCUB improvement** over monolithic designs at iso-node. [STRONG] — anchored to AMD EPYC chiplet architecture (~2× perf/W improvement over equivalent monolithic designs).
- **[CHECK: 2030] In-memory compute will achieve $\geq 10\times$ J/token improvement** over the best digital accelerator on a standard LLM inference benchmark ($\geq$ 7B parameters) at iso-quality. [WEAK] — anchored to single-vendor demonstrations (Mythic, ~5× for vision); not yet replicated in published study.
- **[CHECK: 2030] The JPCUB gap between the most-efficient and median computing platform for AI inference will be $\geq 50\times$** (from approximately 10× in 2025). [STRONG] — anchored to GPU→TPU efficiency gap trajectory (2016: ~30×).

- **[CHECK: 2032] No post-silicon logic device** (spintronic, photonic, or other) will have demonstrated a complete, cascaded logic path (gain >1, fan-out >2, room temperature) with switching energy below 1 fJ. [WEAK] — negative prediction; anchored to general post-silicon device maturity base rates (~0.15 success within 15 years).
- **[CHECK: 2035] Neuromorphic hardware will not have achieved $\geq$ 5% of AI training compute cycles** (inference-only adoption is a separate, weaker claim). [STRONG] — anchored to the historical absence of non-floating-point training and the lack of a published roadmap for spiking backpropagation at scale.
- **[CHECK: 2030] At least one JPCUB-relevant prediction from this register will have been disconfirmed.** [STRONG] — meta-forecast anchored to published forecasting literature (Tetlock and Gardner 2015). A 0/N accuracy would invalidate the method; a mixed record is the expected outcome.

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7. Discussion

7.1 JPCUB as a Leading Indicator: Confirmed with Caveats

The retrospective analysis confirms the hypothesis with three important qualifications. First, JPCUB is a leading indicator *only* in eras where energy efficiency is the primary competitive axis — which defines the post-Dennard era but did not define earlier transitions. Second, JPCUB's lead over paradigm transition is measurable (3-5 years in recent transitions) but not indefinite — JPCUB provides advance warning, not clairvoyance. Third, JPCUB is most informative when comparing within a workload class; cross-workload comparison requires the benefit measure (denominator of JPCUB) to be carefully normalized.

7.2 Heterogeneous Integration as the Winning Strategy

A key finding — one that emerged from the analysis rather than being assumed at the outset — is that every post-silicon candidate (except continued CMOS scaling itself) depends on silicon CMOS as a substrate. In-memory computing uses CMOS for readout and control. Neuromorphic processors use CMOS for spike routing. Spintronic logic requires CMOS for readout circuitry. The most realistic scenario is not "CMOS replacement" but "CMOS + X" — heterogeneous integration where the right computation is placed on the right substrate. JPCUB optimization under this model means reducing data movement between substrates, which is precisely what chiplet integration and in-memory computing do best.

7.3 Limitations

This analysis has several limitations. The retrospective for early transitions (T1-T2) relies on order-of-magnitude estimates from device physics rather than precise benchmark data — the archival record for 1940s-1950s computing is sparse. The prospective probability judgments, while anchored to reference classes and sensitivity-tested, are structured judgments, not empirically derived quantities. The forecast considers seven candidates but cannot rule out a completely unexpected paradigm — reversible computing, quantum-classical hybrids operating on entirely different principles, or a paradigm that does not yet have a published research literature. The calibration register mitigates this risk by including a self-disconfirmation prediction.

7.4 Counterfactual Technology Stacks

A structured backcasting exercise reveals actionable near-term opportunities. If a foundry-neutral memristor reliability consortium had been formed in 2008 (following HP Labs' memristor announcement), logic-grade ReRAM might have been available by 2020 — advancing the in-memory computing timeline by 5-10 years. The most impactful near-term action is building JPCUB benchmarking infrastructure *now*, before the post-silicon transition is complete. The retrospective shows that benchmarking infrastructure consistently lags paradigm adoption by 5-10 years; building JPCUB benchmarks for chiplets, in-memory, and neuromorphic hardware in 2026-2027 would enable JPCUB to function as the leading indicator it is theoretically capable of being.

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8. Conclusion

JPCUB validates as a leading indicator of computing paradigm shifts. Across six historical transitions, the metric transitions from LAG (early eras, when energy was not the primary adoption driver) to LEAD (post-Dennard eras, where energy efficiency determines architectural winners). In the current computing landscape, JPCUB points toward heterogeneous integration — chiplet-based architectures that reduce data movement energy — as the most probable near-term JPCUB improvement path, with in-memory computing as the highest-upside (but higher-risk) alternative.

The most important finding is not the specific ranking of candidates but the structural insight: architecture-level JPCUB improvements (reducing data movement) now dominate device-level improvements (improving switching energy). This contradicts the historical pattern where device improvements — vacuum tube → transistor → CMOS — delivered the largest efficiency gains. In the post-Dennard era, the energy cost of moving data has become the binding constraint, and the paradigms that reduce data movement are the paradigms that will define the next era of computing.

Declarations

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This research was conducted by the QNFO Research Collective without external funding.

Conflicts of Interest

The author is affiliated with QWAV, a commercial entity whose strategy involves JPCUB as a core metric. This paper is a validation study, not a promotional document. The methodology, data, and analysis are presented for independent verification regardless of commercial outcome.

Ethics Approval

Not applicable — this research involves no human subjects, animal subjects, or sensitive data.

Consent to Participate / Consent for Publication

Not applicable.

Author Contributions

Sole author: conceptualization, methodology, data collection, analysis, writing.

Data Availability

All data supporting this analysis is available in the project repository at https://github.com/QNFO/jpcub-validation, including the JPCUB historical dataset (`artifacts/jpcub-historical-data.csv`), the structured forecast protocol (`artifacts/structured-forecast-protocol-v2.md`), the practical applications extension (`artifacts/practical-applications-extension.md`), and the counterfactual backcasting analysis (`artifacts/counterfactual-backcasting.md`).

Code Availability

The JPCUB computation script used to generate the historical dataset is archived in the project repository.

Materials Availability

Not applicable — no physical materials were used.

Use of Artificial Intelligence

The structured forecast assessment in Section 4 involved a same-model consistency check (the agent's probability judgments were independently assessed by a second instance of the same underlying model). This is disclosed as a structured second-opinion exercise, not as independent inter-rater reliability in a statistical sense.

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